

PAPERFLOW

Project Documentation

Editorial workflow management for academic conferences

Purpose

A practical reference for conference staff, operational teams, and technical maintainers.

Scope

Workflow, roles, author portal, storage, email, monitoring, exports, and deployment.

Prepared for the Paperflow project

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1. Product Overview

Paperflow is a Laravel-based editorial workflow application for academic conferences. It brings submission intake, staff assignments, formatting review, author revisions, PDF eXpress handling, EDAS handoff, monitoring, and reporting into one role-aware system.

Core design principle

Every submission transition is validated through the centralized workflow service. Staff permissions, conference membership, and paper-level policies are enforced server-side.

Documentation Map

Section	Use this section for
Roles and access	Understanding who can perform each task.
Workflow	Following a manuscript from submission to completion.
Author Portal	Supporting authors through revisions and final checks.
Technical operations	Maintaining storage, email queues, monitoring, and deployment.

Primary Interfaces

- **Staff workspace:** dashboard, paper list, paper detail pages, conference settings, reconciliation, monitoring, and exports.
- **Public submission:** a conference-specific landing page and form guarded by rate limiting and optional Turnstile verification.
- **Author Portal:** a private, tokenized page for submission details, feedback, file downloads, and requested revisions.

2. Roles and Access

Staff sign in through the same login page using a username or email. Conference memberships are active only when explicitly configured for that conference.

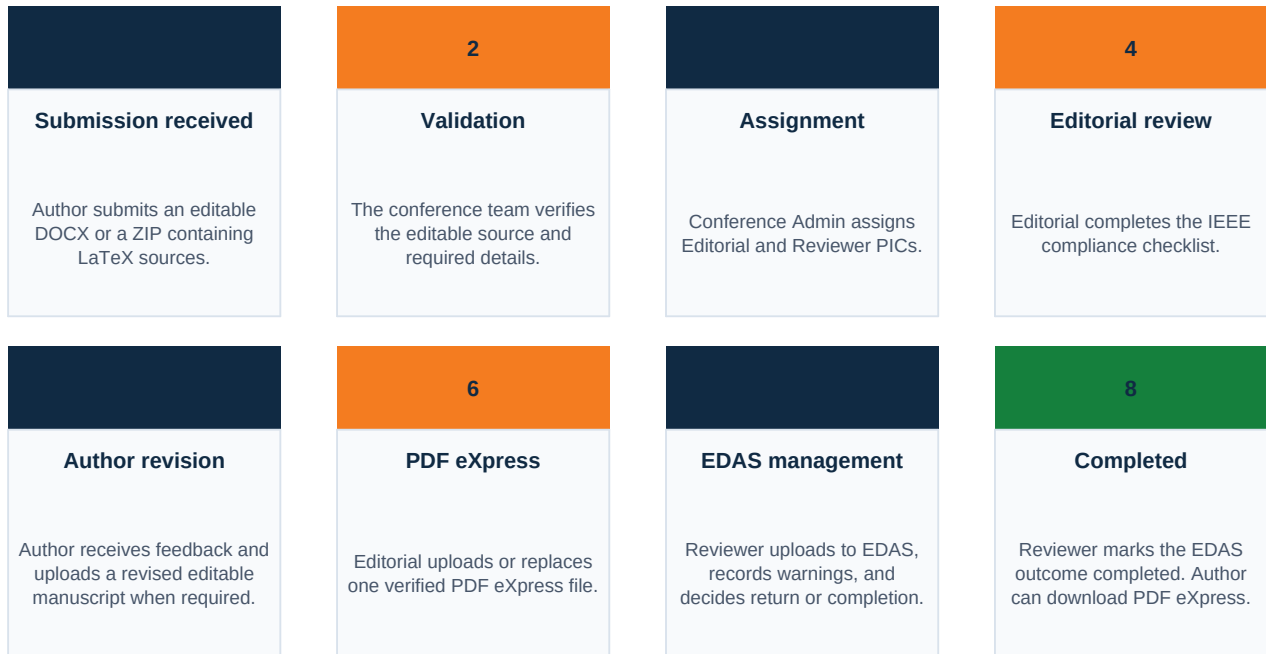
Role	Primary responsibilities	Access boundary
Superadmin	Creates users and conferences, manages system-level monitoring, backups, impersonation, and purge tools.	All data and all conferences.
Conference Admin	Configures conferences, forms, team members, checklists, branding, templates, storage, and assignments.	The conferences they administer.
Editorial	Works on editorial review, compliance checklist, author revisions, and PDF eXpress upload.	Assigned paper operations; conference reporting access where enabled.
Reviewer	Receives the PDF eXpress file, uploads to EDAS, records structured warnings, returns corrections, or completes the paper.	Assigned review operations; conference reporting access where enabled.
Viewer	Reads conference progress and papers without changing workflow data.	Read-only conference access.

Authorization note

Permissions are not based on buttons being hidden. Every important route is protected by policies and active conference membership checks.

3. Submission Workflow

Paperflow keeps workflow transitions explicit. A paper must not be moved by directly updating its status; the workflow service records history, notifications, and audit data.



Key workflow rules

- The author may be asked to revise multiple times. Each revision returns the paper to editorial review and clears the previous revision deadline.
- The final page count and a PDF eXpress file are mandatory before Editorial can send the paper to Reviewer.
- EDAS warning entries are structured internal records. They can be used to return a paper to Editorial or retained as evaluation evidence when the paper is completed.

4. Author Portal

Each submission has a private tokenized portal. It lets authors see the current status, update allowed details, read author-facing feedback, download the relevant manuscript version, and upload a revision only when requested.

Author Revision Safeguards

- The portal shows the latest editable manuscript uploaded by the Editorial Team when one exists.
- Authors must explicitly confirm they used that editorial version as the basis for their revision and that requested corrections have been addressed.
- The server revalidates the selected editorial base file so an author cannot submit against an older version if a newer editorial version exists.
- Author revision uploads are restricted to DOCX or ZIP packages containing LaTeX source files.

Author guidance

The built-in "How the Process Works" modal explains submission receipt, editable-file validation, editorial review, revision cycles, PDF eXpress and EDAS processing, and the final EDAS verification step.

File Naming and Download Behavior

Manuscript downloads are forced as attachments. The browser downloads the file directly with a predictable name: **{paperid}-v{version}-{source}[-guidance].{extension}**. PDF eXpress downloads use **{paperid}-pdf-express.pdf**.

5. PDF eXpress and EDAS

PDF eXpress is handled as one replaceable submission-level file, not as part of editable manuscript version history. Re-uploading replaces the active object and removes the previous storage object to avoid unnecessary storage growth.

Stage	Responsibility	What is recorded
PDF eXpress upload	Editorial	Verified PDF, upload timestamp, uploader, status, and storage metadata.
EDAS upload	Reviewer	EDAS upload time and reviewer action.
EDAS warnings	Reviewer	A structured list of EDAS warning messages for internal follow-up.
Completion or return	Reviewer	Return for editorial correction or mark the EDAS result completed.

- The PDF eXpress file is visible to authorized staff at all workflow stages where appropriate.
- Authors see the PDF eXpress download only after the Reviewer marks the paper Completed.
- EDAS management controls are read-only and server-rejected outside the active Reviewer stage.

6. Technical Architecture

Paperflow uses a server-rendered Laravel architecture. Browser clients do not receive storage secrets, Google OAuth credentials, database credentials, or Gmail app passwords.

Layer	Technology	Purpose
Application	Laravel 12, PHP 8.2+	Workflow rules, policy enforcement, forms, controllers, jobs, and storage routing.
Presentation	Blade, Alpine.js, Tailwind CSS 4, Vite	Responsive staff workspace and Author Portal interfaces.
Database	PostgreSQL 17 on Supabase	Server-only application data through the Supabase Session Pooler.
Files	Private Supabase Storage or Google Drive	Conference-specific file storage selected by each conference.
Mail and queues	Gmail SMTP and Laravel database queue	Branded queued notifications, test sends, retries, and monitoring.

Security Controls

- Policy and membership checks protect staff routes and conference data.
- Author Portal access requires a cryptographically generated, expiring token.
- Public submission uses rate limits, MIME and extension validation, and optional Cloudflare Turnstile verification.
- Supabase buckets remain private. Download URLs are signed or proxied after authorization.

7. File Storage, Email, and Queues

Each conference chooses a storage provider. New uploads follow that setting, while previously uploaded versions remain with their original provider. A conference can therefore contain historical files in both providers.

Storage behavior

Provider	Behavior
Private Supabase Storage	Files are stored in a private bucket. Paperflow uses short-lived signed URLs and forced attachment naming.
Google Drive	Files are stored in the conference-specific authorized folder. Paperflow proxies downloads rather than redirecting to a public Drive view URL.
Failed uploads	For existing submissions, a failed upload can be retained in private local temporary storage and retried without selecting the file again.

Email and queue behavior

- ConferenceMailer creates branded HTML and plain-text email, persists an email log, then dispatches a queued SendLoggedEmail job.
- Workflow messages include assignments, revision requests, reviewer handoff, EDAS return/completion, completed-paper revert, and author revision upload.
- On shared Hostinger hosting, the queue worker is started by a Custom Cron Job calling scripts/paperflow-queue-worker.sh. It uses flock and --stop-when-empty.
- After mail or queue configuration changes, run `php artisan optimize:clear` and `php artisan queue:restart` before starting or allowing the worker to run again.

8. Operations, Monitoring, and Reporting

Paperflow provides operational monitoring for the database, storage, failed jobs, Laravel errors, audit logs, email logs, and scheduled deadline reminders.

Capability	Operational purpose
Email Delivery Monitoring	Review queued, sent, failed, and cancelled messages. Failed messages with stored content can be resent.
Deadline Reminder Monitoring	Review planned and dispatched author/editor revision reminders and their scheduled times.
EDAS CSV Reconciliation	Import EDAS paper ID, title, and author data; cross-check it against Paperflow; inspect warnings and export reports.
Conference Data Export	Create CSV, XLSX, or print-ready PDF reports using selectable paper, author, workflow, and EDAS fields.
Audit Log	Record significant actions, including workflow events, reconciliation, and conference data exports.

Export scope

The conference Data Export submenu is available to active conference staff and exports the complete dataset for that conference. It is always scoped by the conference URL and cannot include another conference.

9. Maintenance and Troubleshooting

Use the following checklist when operating or diagnosing Paperflow. Do not expose secrets in logs, browser code, screenshots, commits, or documentation.

Scenario	Recommended check
Queued email remains queued	Confirm the database queue worker is running. On Hostinger, verify the Custom Cron Job calls the queue-worker shell script at the intended interval.
Author cannot submit a revision	Check status is Waiting for Author Revision or Author Correction Required, file type is DOCX/ZIP, and confirmation fields are present. Check the displayed form error and Laravel log.
File download opens in browser	Use Paperflow download endpoints, not a raw storage URL. They set Content-Disposition: attachment and enforce the standard filename.
Storage upload error	Check the conference storage provider, private bucket credentials or Google Drive folder authorization, then inspect upload attempts and Laravel logs.
EDAS reconciliation mismatch	Confirm the CSV headers and IDs. Paperflow tolerates formatting variations but flags significant ID or title differences.

Standard verification for code changes

- `vendor/bin/pint --dirty`
- `php artisan test`
- `php artisan view:cache`
- `pnpm build`
- `php artisan migrate:status`

Final operational reminder

Database migrations are the source of truth. Never change a workflow status directly when the transition must be validated by SubmissionWorkflow.